Algorithmic Optimization of Dense Sphere Packings in Non-Euclidean Spaces

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I. RESEARCH PLAN

Research Title:

Algorithmic Optimization of Dense Sphere Packings in Non-Euclidean Spaces

Research Topic:

Optimization algorithms for dense packings of points and disks on the sphere and in the Euclidean and hyperbolic planes

Research Problem:

It is not yet clear how gradient descent, simulated annealing, and compression compare under an equal computational budget in packing points on the sphere and disks in balls of the sphere, the Euclidean plane, and the hyperbolic plane, or whether curvature changes the ranking of the methods.

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Algorithmic Optimization of Dense Sphere Packings in Non-Euclidean Spaces

Rationale

This section explains the problem, the existing knowledge on which the study builds, the gap it addresses, and its significance.

Background

Packing equal disks or points as densely as possible is a classical problem of geometry. On the unit sphere, the Tammes problem asks for N points whose smallest pairwise geodesic distance is as large as possible. The problem is named for Tammes (1930). Optimal configurations have been proved for N = 3, 4, 6, and 12 (Fejes Tóth, 1943), N = 5, 7, 8, and 9 (Schütte & van der Waerden, 1951), N = 24 (Robinson, 1961), N = 10 and 11 (Danzer, 1963), N = 13 (Musin & Tarasov, 2012), and N = 14 (Musin & Tarasov, 2015); for most other N, the best known configurations come from computer search (Sloane et al., n.d.; Cohn, n.d.) and are not proven optimal. Methods for spreading many points over a sphere are surveyed by Saff and Kuijlaars (1997).

Curvature changes the geometry of packing. In the Euclidean plane the area of a disk grows as the square of its radius; in the hyperbolic plane it grows exponentially (Cannon et al., 1997), so most of the area of a large hyperbolic disk lies near its boundary. Böröczky (1978) proved density bounds for packings in spaces of constant curvature, where the definition of density is itself delicate. Computational methods for packing include energy minimization, Monte Carlo and annealing methods (Kirkpatrick et al., 1983; Metropolis et al., 1953), and compression of growing disks (Lubachevsky & Stillinger, 1990).

Research Gap, Importance, and Societal Impact

Established knowledge in this area includes the optimal Tammes configurations listed above, the area formulas of the three plane geometries, and the existence of the three families of methods. The proposed contribution of this project is narrower: a transparent, reproducible

comparison of three simple optimizers under one shared protocol, with an equal computational budget and paired random seeds, on one bounded disk-packing problem posed identically in geodesic balls of the sphere (curvature +1), the Euclidean plane (0), and the hyperbolic plane (−1), together with the Tammes problem on the whole sphere. I did not find a study that compares these methods this way. The project proves no theorem and does not aim to set packing records.

Well-spread point sets are used to sample spheres and to place sensors or antennas over curved regions, and hyperbolic packings relate to embeddings of tree-like networks. A reproducible comparison can help students and researchers choose a method and see where each method fails.

Research Questions

The study addresses four questions. RQ1: under an equal budget, which of gradient descent, simulated annealing, and compression reaches the best packings in each problem? RQ2: how does performance change as the number of points or disks grows? RQ3: for numbers of points with a proven optimum, how often does each method recover that optimum within a stated tolerance? RQ4 (exploratory): when the same bounded-disk problem is solved in spaces of curvature +1, 0, and −1, does the ranking of the methods change? This question is exploratory because curvature and area are confounded (a ball of the same radius has a different area in each space), only six settings per space are run, and the ball of radius 3 on the sphere covers nearly the whole sphere.

Goals, Expected Outcomes, and Hypotheses

This section lists the objectives, states the hypotheses, and fixes the rule that decides whether each hypothesis is supported.

Goals and Expected Outcomes

The objectives are (O1) to implement and validate the geometry and the three optimizers from scratch, (O2) to run 30 paired seeds in each of 33 conditions under an equal budget, (O3) to report all results, including failures, and (O4) to release code and data so that every table and figure can be regenerated. The expected outcome is a reproducible package that states which method performs best in which setting, with the size and the limits of each difference. No winner is assumed, and negative results will be reported.

Hypotheses and Decision Rules

Three hypotheses are tested, each with a decision rule stated in Table 1. This plan was finalized after some results had been seen (see Appendix A, History of Work Done, which states which rules were fixed before, and which after, results were seen), so every hypothesis result is to be reported as exploratory, and H1 most of all. The wording of H1 refers to the space, not to curvature alone, because only the bounded-disk problem is posed identically in all three spaces. Table 1 gives the rules.

Table 1

Hypotheses and Decision Rules

Hypothesis	Test and measurable criterion	Decision
H1 (exploratory): the ranking of the algorithms depends on the space	Common formulation (disks in a geodesic ball, six settings per space): Kruskal–Wallis test across the three spaces of the per-setting mean paired difference in relative score, for each of	Supported: some pair has adjusted $p < .05$ and the lowest-mean-rank algorithm is not the same in all three spaces. Partly supported:
	the three algorithm pairs, Holm-adjusted over the three pairs; and the algorithm with the lowest mean rank in each space.	exactly one of the two. Not supported: neither.
H2: compression scales better with N	Tammes problem, $N \geq 13$: per-seed slope of relative score on $\log_{10} N$; Wilcoxon signed-rank tests of the compression slope against each other slope, Holm-adjusted over two tests.	Supported: compression slope higher than both, adjusted $p < .05$. Partly supported: higher than one. Not supported: otherwise,

Hypothesis	Test and measurable criterion	Decision
		including an opposite direction.
H3: every algorithm recovers the proven optimum for small *N*	Tammes problem, *N* = 4 to 14: share of seeds with relative score ≥ .999 (0.1% tolerance) at each *N*.	Supported: share ≥ .90 at every *N* for all three algorithms. Partly supported: for one or two algorithms. Not supported: for none.

Note. Relative score = final score divided by the reference value (see Experimental Conditions). A 1% tolerance (relative score ≥ .99) is reported as a sensitivity check only.

Procedures

This section defines the spaces, problems, experimental conditions, algorithms, metrics, and validation tests. The parameter values given are tunable defaults, not requirements; the ranges examined in the sensitivity analysis are stated in the Data Analysis section.

Spaces and Problems

Three constant-curvature planes are used. The unit sphere S^2 has curvature +1, with points as unit vectors $\boldsymbol{x}$ in R^3 and geodesic distance given by Equation 1. The hyperbolic plane H^2 has curvature −1; computations use the hyperboloid model, where a point satisfies $[\boldsymbol{x}, \boldsymbol{x}] = -1$ and $x_0 > 0$ for the Lorentz product $[\boldsymbol{x}, \boldsymbol{y}] = -x_0y_0 + x_1y_1 + x_2y_2$, and distance is given by Equation 2; figures use the Poincaré disk. The Euclidean plane R^2 has the usual distance.

$$\cos d = \mathbf{x} \cdot \mathbf{y} \tag{1}$$

$$\cosh d = -[\mathbf{x}, \mathbf{y}] \tag{2}$$

Problem A (Tammes) places *N* points on S^2 to maximize the smallest pairwise distance. Problem B places *n* equal disks in a closed geodesic ball of radius *R* about a fixed center, in each of the three spaces (a spherical cap, a Euclidean disk, and a hyperbolic disk), and maximizes the common radius *r*. A disk of radius *r* whose center is at distance ρ_i from the

ball center lies inside the ball when $\rho_i \le R - r$, and two disks do not overlap when $d_{ij} \ge 2r$. Both problems maximize $\ell_{\min}$ in Equation 3, where the second term is absent for Problem A; the score is $\ell_{\min}$ for Problem A and $r = \frac{\ell_{\min}}{2}$ for Problem B. Problem B is a maximum-radius formulation for fixed n and R; the number of disks of a fixed radius that fit is derived from it.

$$\ell_{\min} = \min \left\{ \min_{i<j} d_{ij},\ \min_{i} 2(R - \rho_i) \right\} \tag{3}$$

The area of a disk of radius r is $2\pi(1 - \cos r)$ on S^2 ($r \le \pi$), πr^2 on R^2, and $2\pi(\cosh r - 1)$ on H^2. Packing density is n times the area of a disk of radius r, divided by the area of the ball of radius R.

Comparability of the Problems

Problem A and Problem B differ in their objective (a minimum distance against a common radius), in their constraints (no boundary against a boundary), and in scale, so their results are not compared with one another. Statements about curvature rest only on Problem B, which has the same formulation, the same R and n, and the same code in all three spaces. Even then, a ball of radius R has a different area in each space, so a difference between spaces can come from curvature, from area, or from both; the study does not separate them. Problem A is used for the comparison of the algorithms on the sphere, the recovery of proven optima, and scaling with N.

Experimental Conditions

Problem A uses N = 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 24, 30, 40, and 60. Problem B uses the six settings (R, n) = (2, 7), (2, 12), (2, 19), (3, 12), (3, 19), (3, 30) in each of the three spaces. The evaluation seeds are 1 to 30; the initial configuration depends only on the seed and the condition, so the three algorithms start from the same configuration. Pilot runs for parameter tuning use seeds 9001 to 9004, which are never used in the evaluation. The total is 2,970 runs.

The Tammes problem accounts for 1,350 runs (15 values of N × 30 seeds × 3 algorithms), and each space accounts for 540 runs in Problem B (6 settings × 30 seeds × 3 algorithms). The reference value of a condition is the proven optimum for Problem A with N = 4 to 14 and 24 (credited to Fejes Tóth, 1943; Schütte & van der Waerden, 1951; Robinson, 1961; Danzer, 1963; and Musin & Tarasov, 2012, 2015), the published best-known value where one is entered (Sloane et al., n.d.; Cohn, n.d.), and otherwise the highest score found by any run in that condition; the last two are not claimed to be optimal.

Optimization Algorithms

Three algorithms are implemented in Python with numpy. Table 2 gives the default value of every parameter; each is tunable. All three move points along geodesics and keep points on the manifold, and all report the best configuration they visited, scored exactly from its coordinates.

Table 2

Algorithm Parameters

Algorithm and parameter	Value
GD: energy, exponent	$E_p = (1/p) \ln \Sigma\, \ell_k^{-p}$ over pair distances and, in Problem B, doubled wall clearances; p from 2 to 150, geometric in the share of budget used
GD: step, line search	$\eta = \min\left(\frac{m\,\ell_{\min}^2}{p},\ \frac{0.25\,\ell_{\min}}{\max\lvert g\rvert}\right)$; m starts at 0.5, × 1.3 after success (maximum 2), × 0.5 after failure (minimum 0.001); on failure multiply η by 0.4, at most 5 trials
SA: temperature	Geometric from 0.02 to 10^{-5} times the length scale $\sqrt{\frac{\text{area}}{n}}$
SA: move, adaptation	Random geodesic step of length $\delta(0.1 + 0.9u)$, u uniform on (0, 1); δ starts at 0.3 times the length scale; every 100 proposals δ × 1.15 if acceptance > .40, × 0.87 if < .25
SA: point choice	Probability 0.3: an endpoint of the limiting pair or the point nearest the wall; otherwise uniform

Algorithm and parameter	Value
CP: time step, growth	Time step 0.02 times the length scale at unit root-mean-square speed; common radius grows by γ d*t* per step, γ falling linearly from 0.25 to 0.01, capped by half the smallest distance
CP: collisions	Velocity components along the line of centers exchanged if approaching; overlaps removed; wall reflection in Problem B; kinetic energy renormalized each step

Note. GD = gradient descent; SA = simulated annealing; CP = compression. The length scale is $\sqrt{\frac{\text{area}}{n}}$. The compression method is a simplified, time-stepped version of the method of Lubachevsky and Stillinger (1990). The values are defaults used in the reported runs, chosen on pilot seeds; they are not fixed by the design, and the ranges examined later are listed under Supplementary Exploratory Analyses.

Budget, Initialization, and Stopping

Every run receives a budget of 1,500 sweeps, where one sweep is the $\frac{n(n-1)}{2}$ pair-distance evaluations needed to compute all distances once; a gradient step costs one sweep plus one sweep per line-search trial (in Problem B each evaluation also includes the n wall clearances), a simulated-annealing move costs $n - 1$ evaluations, and a compression step costs two sweeps. Gradient descent and compression stop when their next step would exceed the budget; simulated annealing fixes its number of moves from the budget in advance and does not charge proposals rejected for lying outside the ball, so it may use slightly less than the budget. Initial points are uniform on S^2 for Problem A and uniform by area in the ball of radius $R - 0.05$ for Problem B.

Evaluation Metrics

The primary metric is the relative score, the final score divided by the reference value. A run succeeds if its relative score is at least .999; success rates are reported with a 1% sensitivity check (relative score at least .99). Convergence is the number of sweeps used when the best score first reached 99% of the run's own final score. Scaling is the per-seed

slope of relative score on $\log_{10} N$ for $N \geq 13$, and the log–log slope of mean run time on N. Computational cost is the wall-clock time, reported as descriptive information because the budget is fixed in distance evaluations. Mathematical optimality is claimed only where a published proof exists; elsewhere a result is called the best score found.

Final Round: Design, Rules, and Timestamp

A last round is planned to answer the one question that earlier rounds left open, whether curvature changes the density deficit once the boundary share is in the model, and to test one original rule. Its hypotheses, decision rules, seeds, and analysis code are fixed in PREREGISTRATION_ROUND3.md before any run; Table 3 gives its decision rules and Table 4 its design. Everything that happened in earlier rounds is in Appendix A.

Timestamp. The preregistration file has the SHA-256 hash e9d6cf21391fbd2f262e192940c07482f82e43c08857c58f1624d405eb448424, recorded by the author on 2026-10-30 at 03:43:32 UTC. A hash recorded by the author is not an independent time stamp. Before any run of the final round, the author will email the preregistration file, this plan, and the earlier preregistration files to the research adviser, and the date of that email is the time stamp; no run is started before it is sent. Any deviation will be listed in the Research Paper. This plan was finished before the Research Paper, and its body will not be changed after the final round has run; later events are recorded in the paper.

Table 3

Decision Rules of the Final Round

Question	Decision rule
A1: curvature adds a deficit at a given boundary share	Fit $\delta = a + \beta\, b + g_{\mathrm{H}}$ [hyperbolic] + g_{S} [cap]; the 95% interval of g_{H} is from 10,000 bootstrap resamples of the 48 cells. Detectable if the interval excludes 0 and $g_{\mathrm{H}} \geq .02$; no detectable effect if the interval lies within ±.02; otherwise inconclusive. The same rule is applied to the cap term.
A2: paired differences (descriptive)	Mean difference in δ between the hyperbolic disk (and the cap) and the Euclidean disk over the 48 cells, with a bootstrap interval.
B1: the number of disks can be predicted	$K_{\mathrm{pred}} = \lfloor n_{\mathrm{ref}}(1 - 0.46\, b_0) + 0.5 \rfloor$, with n_{ref} the number of disks that would fit at the upper bound on the density and b_0 the design boundary share. Supported if at least 80% of conditions have $\lvert K - K_{\mathrm{pred}} \rvert \leq 1$, the mean absolute error is at most 1.0 disk, the mean relative error is at most .10, and each space has a mean signed error within ±1 disk; not supported if fewer than 60% are within 1 or the mean absolute error exceeds 1.5; otherwise inconclusive.

Note. The coefficient 0.46 was fitted in an earlier round and is not refitted. With the residual spread of the earlier round, the expected half-width of the interval of g_{H} is about .03, so the round can show an effect of .03 or more but may remain inconclusive; the plan states this before the data. Data: PREREGISTRATION_ROUND3.md.

Table 4

Design of the Final Round

Part	Conditions	Methods, budget, and seeds
A: curvature at equal area	Three spaces (Euclidean disk, hyperbolic disk, spherical cap) × eight areas (2, 3, 4, 5, 6, 8, 10, 12) × six numbers of disks (8, 12, 16, 20, 24, 32) = 144 conditions in 48 cells of equal area and equal *n*.	Polished gradient descent and L-BFGS-B; 1,500 sweeps; fresh seeds 501 to 520; 5,760 runs.
B: a rule for the number of disks	Disk radii 0.25 and 0.35 × four design boundary shares (0.375 to 0.525) in the hyperbolic disk and the cap, and the four boundary shares in the Euclidean disk (no length scale) = 20 conditions.	Polished gradient descent; 1,500 sweeps; fresh seeds 601 to 620; the number of disks *K* is the largest *n* reached in at least 10 of 20 seeds.

Note. The ball radius of each condition follows from its area (Part A) or from the design boundary share and the disk radius (Part B). Data files named in PREREGISTRATION_ROUND3.md.

Validation Tests

The tests in Table 5 are executed before the experiments that depend on the code they check, and those experiments proceed only if every test passes. No test result is reported unless the test was executed.

Table 5

Validation Tests and Pass Criteria

Test group	How it is performed	Pass criterion
Sphere distance	300 pairs vs. arctangent form; closed forms	Error < 10^{-9}; closed forms < 10^{-12}
Hyperbolic distance	Poincaré formula; law of cosines; area integral	Relative error < 10^{-7}; others < 10^{-6}
Euclidean control	Distance of (0,0), (3,4); 7-disk hexagon	Error < 10^{-12}
Spherical cap	Poles, area, sampling, Euclidean limit	Area < 10^{-6}; limit within 0.2%

Test group	How it is performed	Pass criterion
Metric axioms	Symmetry; 4,000 triples per space	Asymmetry $< 10^{-12}$; triangle $< 10^{-9}$
Exponential map	Step length; point stays on manifold	Error $< 10^{-7}$
Gradients	Analytic gradient vs. finite differences	Difference $< 10^{-5}$
Known Tammes optima	Three polyhedra; optimizers, seeds 9101–9105	Gap < 0.002 rad

Note. A failing test blocks the experiments. Recovery by each optimizer is hypothesis H3, not a validation criterion.

Risk and Safety

The project is computational, so its risks are procedural. This section lists approvals and the measures that protect data integrity.

Hazards and Approvals

The project uses no hazardous materials, human participants, or vertebrate animals. It requires a risk assessment: Form 3 (Risk Assessment) is required for all projects, together with Form 1, Form 1A, and Form 1B and this Research Plan. Form 7 is required only if the project continues earlier research.

Disclosure of AI Assistance and Researcher's Contribution

Claude, an AI assistant developed by Anthropic, was used to assist with the writing of this document, including drafting, grammar checking, and language refinement. Other uses of AI assistance in this project, including its use in programming and analysis, are described in the separate AI Disclosure Agreement submitted to the research adviser and the Science Research Committee. GitHub was used to support code management and access to programming resources. The author's original contributions include the selection of the research problem, the formulation of the research questions, and the decisions regarding the research design. The author personally ran and checked the code, verified the results, and can explain and defend the programming procedures, methods, and findings. The author accepts

full responsibility for the accuracy and integrity of every statement, numerical result, citation, and conclusion presented in this research. The code and data are provided to support reproducibility and independent verification of the results.

Data Integrity, Backups, and Reproducibility

One master script, build_all.py, runs the tests, experiments, statistics, figures, and documents with fixed seeds. Every result is saved to a CSV file; SHA-256 hashes of the delivered files are listed in MANIFEST.sha256 and can be checked with sha256sum. Copies of the code and data are to be kept in at least two separate places. Software versions are listed in the code listing, README.md describes every column of every file, and all results, including failures and negative findings, are retained.

Data Analysis

This section states exactly which observations are compared, which tests are used and why, how their assumptions are checked, and how significance is decided.

Comparisons and Tests

For each condition the observations are the 30 final scores of each algorithm; scores of the same seed come from the same starting configuration and form a block. The Friedman test compares the three algorithms across the 30 blocks, and the Wilcoxon signed-rank test (Wilcoxon, 1945) compares each pair on the 30 within-seed differences. Nonparametric paired tests are chosen because scores are bounded, are tied or clustered at local optima, and are not normally distributed. Effect sizes are Kendall's W for the Friedman test and the matched-pairs rank-biserial correlation with the standardized mean difference d_z for pairs. Means carry 95% confidence intervals based on the t distribution; with 30 seeds these intervals are approximate for non-normal scores and are used descriptively.

Assumptions, Multiplicity, and Significance

The tests assume that blocks are independent, which holds because each seed gives an independent random start. The Wilcoxon test also assumes that paired differences are symmetric about their median; this is checked by the skewness of each set of differences (an absolute value above 1 is flagged), and the shares of zero differences are reported. Tests are corrected with the Holm procedure (Holm, 1979), where one family is all tests of the same kind within one space, and the hypothesis tests are corrected across their own pairs. A result is statistically significant when the Holm-adjusted p is below .05. Practical importance is judged separately: a difference counts as practically important when its mean is at least 0.5% of the reference value, so a significant but smaller difference is reported as such.

Properties of the Algorithms and Supplementary Analyses

Only some properties of the algorithms are proven. Proven: the soft-minimum energy lies between $-\ln \ell_{\min}$ and $-\ln \ell_{\min} + \frac{\ln m}{p}$, so it approaches the true objective as p grows; the gradient is checked against finite differences. Not proven: convergence of any of the three optimizers to a global optimum, and any statement that one is better in general; such statements are empirical and are limited to the conditions run. Supplementary analyses, all exploratory and reported separately in the paper, are planned or have been run for: (a) gradient descent followed by a max-min polishing step (a sequential linear program whose cost is not charged in the sweep count), with wall-clock time reported beside equal-sweep results, because simulated annealing costs much more time per sweep; (b) sensitivity to budget, to the soft-minimum exponent and annealing schedule, and to the starting configuration; (c) an equal-area version of the ball problem and larger balls; and (d) a check of the final scores against an independent computation of the distances. Parameter values named in this plan are defaults; the ranges tried are listed in the paper.

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Appendix A: History of Work Done

This appendix records what was done before the final round, in the order it happened, with the dates and hashes that the author recorded. It is a record of past work, not a plan; the plan is the body of this document. The Research Paper reports the results.

Version History and Timing

The plan was finalized after results existed, so the decision rules in Table 1 are not pre-registered in the strict sense; the dated history below states what was changed, and when, after results had been seen. Only the confirmatory round (version 3), the mini-round (version 4), and the follow-up (version 5) are preregistered, and one decision of the mini-round has been withdrawn (version 5.1), and each preregistration is dated by the hash given below, which the author recorded. The results of earlier versions are never changed by later ones.

Version 1, October 25, 2026. An earlier draft exists. This plan makes no claim that it preceded any experiment.

Version 2, October 26, 2026. States exact design values as executed. By then the 27 conditions of Problem A and of Problem B in H^2 and R^2 had been run, and the spherical-cap results had been seen. To the author's knowledge their analysis code was written before their results were inspected; the data package contains no dated record of the order. After those results were seen, the spherical-cap family (six settings) was added to study curvature in one common formulation; its analysis (Table 1, H1) was written into the analysis code before the cap runs were executed, but H1 as now worded is exploratory with respect to the earlier results. The 0.5% practical-importance threshold and the assumption checks were formalized after the first 27 conditions had been run; the compression method was simplified to a time-stepped scheme; the validation criterion for optimizer recovery was changed to the best of the three optimizers after compression failed an earlier per-optimizer version; and the wording of the second H1 criterion was clarified after the cap results were seen, from "differs in at least

two of the three spaces" to "is not the same in all three spaces" (the two readings give the same decision).

Version 2.1, October 28, 2026. Statements about timing were corrected to match the records; the budget description in Procedures was clarified to match the code; and a sentence about work breaks was removed from Risk and Safety. After results were seen: the reference values for Tammes problems at $N \geq 30$ were taken from published best-known values (Sloane's tables, via Cohn's compilation) rather than the best score found by the runs, and the relative scores were recomputed; the citations for the proven optima were corrected to Fejes Tóth, Schütte and van der Waerden, Danzer, Robinson, and Musin and Tarasov; all hypothesis results and RQ4 were labeled exploratory; supplementary exploratory analyses were added (gradient descent with a final polishing step, wall-clock time, sensitivity to budget, parameters and starting configuration, an equal-area comparison and larger balls); and an AI-assistance statement was added. The 27 conditions, the six cap settings, the three optimizers, the budget, and the decision rules in Table 1 were not changed.

Version 3, October 29, 2026. Adds a preregistered confirmatory round (see the next section), which uses fresh seeds, three budget levels, equal-area balls, added baselines, and a hardware record. Its preregistration file has the SHA-256 hash 39bee69da1ffb2a23961dcbfeafafe9e63b421b5f88c9c79c27036cfadeaa7a0, computed on 2026-10-28 at 21:35:04 UTC before the pilot tuning and before any evaluation seed was run; the tuned parameters were hashed before the evaluation run. A hash recorded by the author is not an independent time stamp. No hypothesis, decision rule, or result of versions 1 to 2.1 was changed. Any further change will be recorded here and in the paper.

Version 4, October 30, 2026. After the results of the confirmatory round had been seen, the author recorded the following in a second file (PREREGISTRATION_ADDENDUM_1.md, SHA-256

abfc572a207fdb943b067ff54851cbabe5736039b0ed399c10144b4db87225dd) and a mini-round file (PREREGISTRATION_MINIROUND.md, SHA-256 3b6702f683023611d6f4a7549317d3e11be674fff08a85ed451bf46969df5492), both hashed on 2026-10-29 at 07:32:46 UTC before any run they describe: (1) the Packomania values for the Euclidean conditions were entered after the first analysis, which had used the best score found (the decisions were the same with both references); (2) the candidate ranges of compression and L-BFGS-B were extended, because their selected values lay at the edge of the old ranges, and both methods were run again (new tuned values hashed 2026-10-29 at 22:34:59 UTC, SHA-256 4a8f8b799b5e641b591f7a8b9887ad260c1834d4ccc51b0a57a514de1a747b6b, before the evaluation); (3) the confirmatory round was run again with the same seeds and saved the final configuration of every run, and the wall contact is reported for all runs; (4) a mini-round with a fixed disk radius and a matched boundary share was added (seeds 301 to 320); and (5) the reference density is described as an upper bound on the density, not as the density of a densest packing. The first preregistration file was not edited, and no result of versions 1 to 3 was changed. Because the author recorded the hashes, they are not independent time stamps; the author reports having emailed the four preregistration files to the research adviser on October 30, 2026, after the runs of the confirmatory round and the mini-round and before the runs of the follow-up, and the adviser’s email, not the author’s hash, is the independent record (it cannot show that the first three files preceded their runs).

Version 5, October 30, 2026. Adds a preregistered follow-up to the mini-round (PREREGISTRATION_FOLLOWUP.md, SHA-256 d8a5f5101148869a4a39c91c4e1a64f938a41a2f96caa2224c0b49b281334a0b, hashed on 2026-10-29 at 23:33:06 UTC before its runs). It corrects a flaw of the mini-round: the Euclidean plane has no length scale, so the Euclidean conditions at disk radii 0.30 and 0.40 were the same problem and the six Euclidean conditions were three, which also overstated the degrees of freedom of rule M2. The follow-up uses five design boundary shares, three disk radii for the hyperbolic disk and the cap, one Euclidean condition per boundary share, seeds 401 to 420, and two methods (polished gradient descent and L-BFGS-B) for the number of disks K. No rule, result, or hash of versions 1 to 4 was changed; the mini-round decisions are reported as they were, with a sensitivity analysis without the duplicates.

Version 5.1, October 30, 2026 (a withdrawal). The decision of rule M2 of the mini-round, “no detectable curvature effect” (hyperbolic term 0.007, interval −0.032 to 0.046, 13 degrees of freedom), is withdrawn. The Euclidean conditions at disk radii 0.30 and 0.40 are the same problem, so the 18 conditions were 15 and the degrees of freedom 10; without the duplicates (a post hoc analysis, decided after the result was seen) the interval is −0.058 to 0.060 and the decision is inconclusive. Earlier text, including the paper written before this correction, said that no curvature effect larger than .05 was found in the mini-round; that statement is withdrawn. The correction note (CORRECTION_NOTE_1.md, SHA-256 80df1f48699de18153ca678d31e4758acf98bc8233f502a51d178987f23fe1ff, hashed 2026-10-30 at 02:51:08 UTC, recorded by the author) states this, and the mini-round decisions as originally computed are kept in mini_decisions.json and reported next to the corrected analysis.

The runs of the follow-up began about one minute after the hash of its file was recorded (23:34 UTC on October 29, 2026; run times in FOLLOWUP_RUNTIMES.txt), and

the replacement boundary shares and the second method were in that file; the author reports that the four files had been emailed to the research adviser before the runs began. F1 to F3 are therefore preregistered by the author's hash, and the adviser's email is the independent record of the date of the files.

Version 6, October 30, 2026. The plan was restructured so that its body looks forward: it states the hypotheses, decision rules, seeds, and timestamp of one final round (PREREGISTRATION_ROUND3.md, SHA-256 e9d6cf21391fbd2f262e192940c07482f82e43c08857c58f1624d405eb448424), and the history of earlier work, including every version above, is kept in this appendix. The final round was preregistered to settle the curvature question that three earlier rounds left inconclusive (more equal-area conditions, with the bootstrap taken over cells of equal area and equal number of disks) and to test a rule for the number of disks. No rule, result, or hash of versions 1 to 5.1 was changed.

Confirmatory Round

This section describes a second round of experiments whose hypotheses, thresholds, and decision rules were written and hashed before any of its evaluation seeds was run. The full text is PREREGISTRATION.md in the data package.

Prediction and Design

Curvature and area are confounded in the first study, because a ball of a given radius has a different area in each space. The confirmatory round therefore compares balls of equal area. For n disks of radius r in a ball of radius R, the density deficit is $\delta = 1 - \frac{n\,\mathrm{Area}(r)}{\mathrm{Area}(R)} / \rho^{*}$, where ρ^{*} is an upper bound on the density of any packing of disks of radius r in the whole space (Fejes Tóth, 1943; Böröczky, 1978), and the boundary share is $b = 1 - \frac{\mathrm{Area}(R-r)}{\mathrm{Area}(R)}$. The prediction is that δ increases with b in every space, so that most of the deficit is a boundary effect and not a curvature effect.

The design uses fresh evaluation seeds 101 to 120, three budget levels of 500, 1,500, and 3,000 sweeps as part of the main design, and six methods: gradient descent, gradient descent with a polishing step, simulated annealing, compression, SciPy L-BFGS-B (Virtanen et al., 2020; Byrd et al., 1995), and basin hopping (Wales & Doye, 1997). Each method has one tuned parameter with six candidate settings, tuned on the same pilot conditions and pilot seeds 9101 to 9104. The 33 conditions are the Tammes problem for N = 9, 13, 14, 24, 30, 40, and 60; balls of equal area 3, 6, and 12 with 12 and 24 disks in the spherical cap, the Euclidean disk, and the hyperbolic disk; and hyperbolic balls of radius 2, 3, 6, and 8 with 24, 36, 50, and 70 disks, each paired with a Euclidean disk of the same area. Equal area in all three spaces is possible only up to 4π, so the large balls compare two spaces. The parameter values are tunable defaults, not requirements. The hardware and software are recorded before the run, and no run is excluded.

Reference values are the proven optimum or the published best-known value (Sloane's tables) for the Tammes problem, the Packomania value (Specht, n.d.) for the ten Euclidean conditions with 12, 24, 36, 50, or 70 disks (entered after the first analysis), and otherwise the best score found in the round, which is stated wherever it is used. As an original element, a count-based version of the disk problem asks for the largest number of disks of a fixed radius that fit in a ball of a given area; it is exploratory and has no decision rule.

Decision Rules

Table 6

Decision Rules of the Confirmatory Round

Question	Decision rule (1,500 sweeps unless stated)
C1: the density deficit follows the boundary share (26 ball conditions)	Supported if Spearman r_s between b and δ is at least .70 and the one-sided permutation p is below .05; not supported if r_s is below .30; otherwise inconclusive.
C2: curvature adds a deficit beyond the boundary share	Least-squares fit of δ on b and indicators of space; a detectable effect if the 95% bootstrap interval of the hyperbolic term excludes 0 and the term is at least .02; no effect if the whole interval lies within ±.02.
C3: the ranking of gradient descent and annealing depends on the space (18 equal-area conditions)	Dependent if the Kruskal–Wallis p is below .05; independent if each 90% bootstrap interval of a difference of medians lies within ±.01.
C4: stronger baselines matter; polished gradient descent is competitive	L-BFGS-B or basin hopping beats both gradient descent and annealing in at least half of the 33 conditions; polished gradient descent ranks first or second in at least half.
C5: the ranking of the methods is stable across budgets	Median Kendall's W over the three budget levels is at least .80.
C6: some method recovers the proven optimum for N = 9, 13, 14, 24	The best share of seeds with relative score at least .999 is at least .90 for all four.

Note. All six questions are separate, are reported whatever the outcome, and are not corrected for one another. Analyses not listed here are exploratory. Data: PREREGISTRATION.md.

Mini-Round With a Matched Boundary Share

The main round could not separate curvature from the boundary effect (C2 was inconclusive), and its boundary share depends on the radius each method finds. A mini-round fixes the disk radius r_0 (0.30 or 0.40) and a design boundary share b_0 (0.40, 0.45, or 0.50) and chooses the radius of each ball so that b_0 is the same in the Euclidean disk, the hyperbolic disk, and the spherical cap (18 conditions). For each ball, K is the largest number of disks for which polished gradient descent at 1,500 sweeps reaches radius r_0 in at least 10 of 20 fresh

seeds (301 to 320), and the fixed-radius deficit is $\delta_0 = 1 - \frac{K\,\mathrm{Area}(r_0)}{\mathrm{Area}(R)\,\rho^*}$. The full text is PREREGISTRATION_MINIROUND.md.

Table 7

Decision Rules of the Mini-Round

Question	Decision rule
M1: the fixed-radius deficit follows the design boundary share	Supported if Spearman r_s between b_0 and δ_0 is at least .50 and the one-sided permutation p (20,000 permutations) is below .05; not supported if r_s is below .20; otherwise inconclusive.
M2: curvature adds a deficit at a matched boundary share	Least-squares fit of δ_0 on b_0, indicators of the hyperbolic disk and the cap, and r_0 (13 degrees of freedom); a detectable effect if the 95% t interval of the hyperbolic term excludes 0 and the term is at least .05 in size; no detectable effect if the whole interval lies within ±.05; otherwise inconclusive. The preregistered fit counted 13 degrees of freedom, which included three duplicated Euclidean conditions; the decision it produced is withdrawn (Version 5.1).
M3: a boundary-loss model describes the other spaces	$\delta_0 = \lambda\, b_0$, with λ fitted on the six Euclidean conditions and not refitted; the model describes the other twelve conditions if their mean absolute error is at most .05, and does not if it exceeds .10; otherwise inconclusive.

Note. The three rules are separate and are not corrected for the six rules of the main round. Analyses not listed here are exploratory. The decision of M2 is withdrawn (Version 5.1). Data: PREREGISTRATION_MINIROUND.md.

Follow-Up to the Mini-Round

The mini-round was inconclusive for M1, and its Euclidean conditions at two disk radii were the same problem. The follow-up uses five design boundary shares (0.35 to 0.55), three disk radii (0.20, 0.30, and 0.40) for the hyperbolic disk and the cap, one Euclidean condition per boundary share, and fresh seeds 401 to 420 (33 conditions). The number of disks K is the larger of two lower bounds, from polished gradient descent and from L-BFGS-B.

Tables 6, 7, and 8 give the decision rules of the confirmatory round, the mini-round, and the follow-up. Analyses outside these rules, including the count-based problem and every robustness check, are exploratory and are labeled so in the paper.

Table 8

Decision Rules of the Follow-Up

Question	Decision rule
F1: the fixed-radius deficit follows the design boundary share (33 conditions)	Same thresholds as M1: supported if Spearman r_s is at least .50 and the one-sided permutation p is below .05; not supported if r_s is below .20; otherwise inconclusive.
F2: curvature adds a deficit at a matched boundary share	Same rule as M2, fitted with indicators of the two radii other than 0.30 (27 degrees of freedom): detectable if the 95% t interval of the hyperbolic term excludes 0 and the term is at least .05; no detectable effect if the interval lies within ±.05; otherwise inconclusive.
F3: a boundary-loss model describes the other spaces	Same rule as M3, with λ fitted on the five Euclidean conditions and used to predict the other 28; at most .05 describes, more than .10 does not. The model is a description, not an explanation.

Note. The three rules are separate and are not corrected for the nine rules of the earlier rounds. Analyses not listed here, including the second method alone and the mini-round without duplicates, are exploratory. Data: PREREGISTRATION_FOLLOWUP.md.